

Trader Performance Analysis using Market Sentiment

Objective

This project investigates how Bitcoin market sentiment influences trader performance using historical trading data and the Fear & Greed Index. The objective is to determine whether market psychology impacts profitability and trading behavior.

Methodology

- Loaded and cleaned two datasets.
- Converted timestamps into daily dates.
- Merged trader data with market sentiment.
- Calculated:
 - Average Closed PnL
 - Win Rate
 - Average Trade Size
 - Trades Per Day
 - Long/Short Ratio
- Segmented traders into:
 - Frequent vs Infrequent
 - Consistent Winners vs Others
 - Large vs Small Position Traders
- Visualized the results using Matplotlib.

Key Insights

1. **Trader profitability is highest during Greed and Extreme Greed periods**, indicating better market conditions for profitable trading.
2. **Win rates increase as market sentiment becomes more positive**, suggesting that bullish markets provide higher probabilities of successful trades.
3. **Average trade size increases during Greed periods**, indicating greater trader confidence and willingness to take larger positions.

Strategy Recommendations

1. **Reduce position size during Fear and Extreme Fear periods** to limit downside risk when market uncertainty is high.

2. **Increase trading activity selectively during Greed and Extreme Greed periods**, where historical data shows stronger profitability and higher win rates.

Conclusion

The analysis demonstrates that market sentiment has a measurable impact on trader behavior and performance. Incorporating sentiment indicators into trading decisions can improve risk management and optimize trading strategies. Because the provided dataset did not include leverage information, trade size was used as a proxy for trading aggressiveness.